Common Divisors

Time limit: 1.00 s **Memory limit:** 512 MB

You are given an array of n positive integers. Your task is to find two integers such that their greatest common divisor is as large as possible.

Input

The first input line has an integer n: the size of the array.

The second line has n integers $x_1, x_2, \dots, x_n$: the contents of the array.

Output

Print the maximum greatest common divisor.

Constraints

- $2 \le n \le 2 \cdot 10^5$
- $1 \le x_i \le 10^6$

Example

Input	Output
5 3 14 15 7 9	7